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ARJO HONG KONG

Early Release — Scope Definition

2-Month Build | 2 Developers | All 3 User Groups

Version 1.0 | May 2026

Purpose of this document: This document defines the agreed scope for the Arjo HK early release. It lists every function that is IN scope (with a clear description), every function that has been deliberately cut (with the reason and target release), the four deliberate cuts made to fit a 2-developer, 2-month timeline, and what each user group will be able to do on day one.

1. Release Context

Parameter	Decision
Target customer	Arjo Hong Kong — medical equipment service operations
Release goal	Replace current paper / phone / Excel process. Parallel running expected at the start while staff gain confidence.
Timeline	2 months from development start
Team size	2 developers working in parallel streams
User groups in scope	All three — Back Office coordinators, Field Engineers, Hospital Staff
Primary workflows	New device installations + Repairs + Preventive maintenance (full mix)
Map provider	Google Maps (international standard)
Mobile apps	Engineer App — native iOS + Android (React Native). Hospital staff — mobile-responsive web form, not a native app.
Smart features	1 beta feature included: Natural Language Service Manual Q&A. Location features are core, not beta.
Scope of this document	IN scope functions + deliberate cuts + day-one user experience

Hospital App decision: Hospital staff use a mobile-responsive web page for fault reporting rather than a native app. This removes approximately 1–2 weeks of native app development and eliminates app store submission. The fault reporting experience is functionally identical — staff tap a link, fill in the form, upload photos, submit. The native Hospital App will be built in G2.

2. Functions In Scope

Every function below is confirmed in scope for the Arjo HK early release. Location features are marked ⊕. The single smart beta feature is marked [Beta].

A Job Management

5 functions

✓ A-01 **Job Creation — All Three Types**

Back Office creates installation, repair, and preventive maintenance jobs. Fields: device, hospital, priority (normal/urgent), scheduled date, job description. Each type has its own workflow path.

✓ A-02 **Job State Machine & Audit Trail**

Strict state transitions enforced by the API: Created → Assigned → In Progress → Completed, with branches for Paused, Resumed, Urgent, Timer Anomaly, Cancelled. Every state change is recorded with actor, timestamp, and notes. Immutable audit log.

✓ A-03 **Manual Engineer Assignment**

Back Office coordinator selects an engineer from the list and sets the scheduled date and time. Engineer receives an immediate push notification on their phone.

✓ A-04 **NFC Scan to Start / End Job** *Core differentiator*

Engineer taps phone to NFC sticker on the device to start the job — timer begins, GPS is locked, hospital geofence activates. Taps again to end — timer stops, GPS confirmed within geofence, service record submitted. The uid read from the tag must match the job's asset_id or the tap is rejected.

✓ A-05 **Job Pause & Resume (Parts Shortage)**

Engineer reports a parts shortage in the app — job enters Paused state automatically. Back Office sees the job in the shortage queue. When parts arrive, Back Office confirms and reschedules. Engineer receives resume notification.

B Asset & Device Management

7 functions

✓ B-01 **Installed Asset Registry — CRUD**

Back Office creates and maintains a record for every physical device installed at an Arjo HK hospital: product model, serial number, hospital, floor and room, installation date, warranty expiry. This record is created before the installation job so the engineer arrives with full context.

✓ B-02 **NFC Tag Commissioning (Write)**
During a new installation, the engineer writes the asset UUID to a blank NFC sticker on the device. The app builds the NDEF payload {uid, v:1} and writes it. System records commissioning time and engineer. Required for all new device installations.

✓ B-03 **NFC Scan to Retrieve Device Info**
Any scan of a commissioned device NFC sticker returns the full device record, current open job, last 3 service records, contract coverage status, and a link to the product spec PDF. Works for both the Engineer App and the hospital web fault form.

✓ B-04 **NFC Tag Replacement Flow**
If a tag is damaged or unreadable, the engineer searches for the asset manually by serial number. Back Office confirms identity remotely. Engineer writes a new blank tag with the same UUID. Replacement event is logged.

✓ B-05 **Product Catalogue**
All Arjo product models pre-loaded before handover. Engineers see model name, standard parts list, spec PDF link, and default PM cycle. Read-only for engineers. Back Office can add or edit models.

✓ B-06 **PM Date Auto-Calculation**
When a maintenance job completes, the system automatically calculates the next PM due date (completion date + product PM cycle in months) and writes it to the asset record. For new installations, the first PM date is set from the installation date.

✓ B-07 **Designated Engineer Assignment**
Each asset can have a designated default engineer. When a maintenance job is created for that asset, the system auto-fills the assigned engineer. Back Office can override at any time.

C Contract Management

3 functions

✓ C-01 **Hospital Contract CRUD**
Back Office creates and maintains service contracts for each Arjo HK hospital: contract number, type (full/partial/emergency only), start and end dates, response SLA hours, account manager, primary contact. Foundation for all billing decisions.

✓ C-02 **Contract Coverage Configuration**
Back Office manually configures which device models and parts each contract covers, flagging each as in-contract (no charge) or out-of-contract (billable). When a service record is submitted, the system automatically applies billing flags — engineers never decide manually.

✓ C-03 **Contract Expiry Warning**

30 days before a contract expires (configurable), the Back Office dashboard shows a warning panel listing contracts nearing expiry. Prevents accidental lapse of coverage.

D Service Manual — PDF & AI Search

2 functions

✓ D-01 **Product Spec PDF Upload & Storage**
Back Office uploads the manufacturer service manual PDF for each Arjo product model. Files stored in S3-compatible storage. Engineers can download and read them in the app. Pre-loaded before handover.

✓ D-03 **Natural Language Service Manual Q&A** ***Beta***
After scanning a device NFC, the engineer can type a question in plain English — e.g. 'How do I replace the sling bar assembly?' or 'What is the safe working load?' The system searches the PDF using vector embeddings and returns the top 3 relevant sections with page numbers and one-tap navigation. Works offline if the PDF has been pre-downloaded.

E Cost Management

3 functions

✓ E-01 **Server-Side Automatic Cost Calculation**
Labour cost is calculated automatically: NFC timer duration in minutes ÷ 60 × engineer hourly rate from their profile. The API never accepts a client-submitted total. All cost data is server-computed.

✓ E-02 **Overhead Expense Logging**
Engineers log mileage (converted automatically at the configured rate per km) and meal expenses (subject to daily cap). Each entry requires a receipt photo. Logged in the app during or after the job.

✓ E-03 **Parts Usage & Auto Billing Determination**
Engineers select parts used from the catalogue and enter quantities. The system automatically checks each part against the hospital contract and generates a classified breakdown: in-contract (absorbed) vs. billable. No manual billing decision by the engineer.

F Engineer Management & Location — Google Maps

5 functions

Location stack: Four location features are included in this release, all built on the same Google Maps + GPS infrastructure. They are core features, not beta. Together they give the Back Office real-time operational visibility that no paper-based system can match.

✓ F-01 **Engineer Profile Management**
Back Office maintains profiles for all Arjo HK engineers: name, employee ID, grade, hourly rate, mileage rate, meal cap, certifications, and assigned regions. These values feed directly into cost calculations.

✓ F-02 **GPS Clock-In / Clock-Out**
Engineers clock in via the app at the start of their working day — this activates GPS tracking. GPS pings are sent every 2 minutes during work hours. Clocking out stops tracking. Location data is retained for 30 days then automatically purged.

⊕ F-03 **Back Office Live Map — Google Maps** ***Location Feature***
The Back Office map shows two live layers simultaneously: (1) All Arjo HK hospital locations as pins — green for active contract, amber for expiring within 30 days, grey for expired. (2) All active engineers as live GPS dots with status colours — green = on-site and working, blue = in transit, yellow = idle/available, red = timer anomaly, grey = off duty. Clicking any engineer dot shows their name, current job, and status. Clicking a hospital pin shows all assets and open jobs at that site.

⊕ F-04 **Hospital Geofence + Timer Anomaly Alert** ***Location Feature***
When an engineer starts a job via NFC tap, a 200-metre geofence is activated around the hospital. If the engineer's GPS shows they have left this boundary while the job timer is still running, a 5-minute countdown starts. After 5 minutes: push notification fires to the engineer ('Still on site? Tap NFC to close the job') and the Back Office receives a timer anomaly alert with the engineer's name, job, and last known location.

⊕ A-07 **Opportunistic PM Alert** ***Location Feature***
When an engineer's GPS confirms they are on-site at a hospital, and that hospital has a device with PM due within N days (configurable in System Config, default 2 days), the Back Office receives a real-time alert: 'Engineer [Name] is at [Hospital] — PM due for [Device] in 2 days. Allocate now?' Back Office can approve to create and assign a PM job immediately, avoiding an extra site visit. Does not auto-create a job — human confirmation required.

G Fault Reporting

3 functions

Hospital staff — mobile web, not native app: Hospital staff access fault reporting through a mobile-responsive web page, not a downloaded app. They open a link on their phone browser. The experience is identical to a native app for this use case — NFC scan to identify the device, form pre-filled, photos uploaded, submitted. No app store install required.

✓ G-01 **Hospital Staff Fault Report — Mobile Web**
Hospital staff open the Utiliti web link on their phone. They tap the NFC sticker on the faulty device — the form auto-fills with device details. They add a fault description, select severity

(low / medium / high / critical), upload photos, and submit. Back Office sees the report instantly.

✓ G-02 **Back Office Manual Fault Entry**
When a hospital phones in a fault, the Back Office coordinator manually creates a fault report: selects the hospital and device, enters the problem description, and can immediately generate a repair job from the same screen.

✓ G-03 **Fault Status Tracking**
Hospital staff can check the status of any submitted fault report via the web page (Received → Engineer Assigned → In Progress → Resolved). No need to call for updates.

H Parts & Inventory

3 functions

✓ H-01 **Parts Catalogue & Inventory Management**
Back Office maintains the Arjo HK parts catalogue: part number, name, unit cost, supplier. Stock quantities are updated manually. When stock falls below the configured minimum, the dashboard shows a low-stock alert.

✓ H-02 **Parts Shortage Report & Job Pause**
Engineer selects the unavailable part in the app and submits a shortage report. The job automatically enters Paused state. The Back Office shortage queue updates in real time, showing which jobs are waiting and for which parts.

✓ H-03 **Parts Arrival Confirmation & Job Resume**
Back Office marks parts as arrived in the system. The system flags all paused jobs waiting for those parts. Back Office can reschedule immediately or set a future date. Engineer receives a job resume push notification.

I Reports

2 functions

✓ I-01 **Core Operational Report**
Back Office can view: total jobs by status, average resolution time by job type, first-fix rate, and jobs completed per engineer — filterable by date range. Export to PDF or CSV.

✓ I-02 **Cost View Per Job**
Each job detail page shows the complete cost breakdown: labour (calculated from NFC timer), overhead expenses (mileage + meals), and parts (in-contract vs. billable). No separate report export at this stage — visible on the job record.

J System Configuration & Notifications

4 functions

✓ J-01 **System Parameter Configuration**

Back Office configures: mileage rate (per km), meal cap (per day), PM advance notice window (default 2 days), geofence radius (default 200m), geofence alert countdown (default 5 minutes), contract expiry warning (default 30 days), Google Maps API key.

✓ J-02 **User & Permission Management**

Back Office manages all Arjo HK system users: back office staff, engineers, and hospital contacts. Assigns roles (admin, coordinator, engineer, hospital_user). Manages engineer profiles (grade, rates, certifications).

✓ J-03 **Push Notifications**

Real-time push notifications sent to the Engineer App (iOS + Android) for: new job assignment, job status changes, parts arrival, geofence alert. Hospital staff receive browser notifications for fault report status changes.

✓ J-04 **WebSocket Real-Time Updates — Back Office**

The Back Office receives live updates without page refresh: job status changes, engineer GPS updates (map dots move in real time), shortage reports, PM due alerts, geofence alerts. Powered by Laravel Echo.

3. Deliberate Cuts — and Why

Four categories of features have been deliberately excluded from this release to make the 2-month, 2-developer timeline achievable. All cut features are logged against their target release.

Cut principle: Every cut was made to protect the core proposition. The NFC job flow, live map, automatic cost calculation, and contract billing are all intact. What was cut is either AI-heavy, needs historical data to be meaningful, or adds an entire app deployment. None of the cuts affect Arjo's ability to run their daily operations on this release.

Cut 1 — Hospital Native App → Mobile Web Page

Instead of building a full native iOS/Android Hospital App, hospital staff access fault reporting through a mobile-responsive web page. The experience for the user is functionally the same — they tap a link, the page opens, they scan the device NFC (via Web NFC API on Android, or manual search on iOS), fill in the form, and submit.

Time saved	~1–2 developer-weeks
What is lost	Background push notifications to hospital staff (they check status by opening the link). iOS NFC auto-fill not available via web (manual device search instead).
Target release	Native Hospital App with full NFC on both platforms → G2
Risk	Low — hospital staff interaction is infrequent. The web form meets the need.

Cut 2 — Natural Language Contract Query → G2

The ability for staff and engineers to ask natural language questions about contract coverage ('Is this part covered under Queen Mary Hospital?') has been cut from this release. Contract coverage is still enforced automatically — the billing determination logic is fully in scope. What is cut is the conversational AI interface on top of it.

Time saved	~1–2 developer-weeks (vector indexing of contract data + conversational API)
What is lost	Staff must check coverage by looking at the contract record in the Back Office. Automatic billing flags still work — engineers are never affected.
Target release	Natural language contract query → G2 (C-05)
Risk	Low — the billing determination is automatic. Manual lookup is needed for unusual queries only.

Cut 3 — Contract PDF Auto-Parsing → G2

When a new contract is uploaded, the AI-powered extraction of key fields (covered models, parts, SLA, expiry date) has been cut. Contract data is entered manually by Back Office coordinators. Given Arjo HK's contract volume at launch, this is manageable.

Time saved	~2 developer-weeks (Python FastAPI microservice + PDF parsing + field extraction model)
What is lost	Manual data entry for contracts instead of AI-assisted extraction. For a small number of contracts this is a 30-minute task per contract.
Target release	Contract PDF auto-parsing → G2 (C-04)
Risk	Low at launch. Would become friction if contract volume grows significantly.

Cut 4 — Detailed Cost Reports & Export → G2

Cost data (labour, parts, expenses) is fully calculated and visible on each job detail page. What is cut is the dedicated multi-dimensional cost report — filterable by engineer, hospital, or contract period with PDF/CSV export. The core I-01 operational report remains in scope.

Time saved	~0.5–1 developer-week (report builder, filter logic, export formatting)
What is lost	No aggregated cost report export at launch. Costs can be viewed per job. Finance team would need to collate manually for the first period.
Target release	Multi-dimensional cost reports with export → G2 (E-04, I-02 enhanced)
Risk	Low for a pilot. Arjo HK's finance team can work from job-level data for the first 2–3 months.

All Other Features Deferred

The following features from the full MVP and G2/G3 roadmap are out of scope for this release and are not listed as cuts — they were never planned for the Arjo HK early release.

Feature	Release
A-06 Location-aware engineer scheduling recommendation	G2
A-08 AI batch scheduling optimisation	G3
B-08 Full lifecycle cost tracking per asset	G2
B-09 Asset health scoring	G3
C-06 Contract profitability analysis	G3
D-04 On-site fault diagnosis guidance	G3

E-05 Expense approval workflow	G2
E-06 Invoice export & finance system integration	G2
F-05 Engineer performance dashboard	G2
F-06 GPS proximity sort for engineer recommendation	G2
G-04 SLA response time breach alert	G2
H-04 Parts usage & cost analytics	G2
H-05 AI predictive parts reorder	G3
I-03 Fault frequency heat map	G2
I-04 Equipment fault pattern analysis	G3
I-05 Contract profitability report	G3
J-05 Email / SMS notifications	G2
All G3 AI features	G3

4. What Each User Gets on Day One

This is what the three Arjo HK user groups can do from the moment the system goes live.

🖥️ Back Office Coordinator

- Create and manage all job types — installation, repair, PM
- See all engineers on a live Google Map — who is where, what they are doing, real-time
- Hospital pins show contract status at a glance — green, amber, or grey
- Receive geofence alerts if an engineer leaves a hospital while a job timer is running
- Get PM opportunity alerts when an engineer is already on site
- Handle parts shortages — see the queue, confirm arrivals, reschedule
- View cost breakdown on every completed job — no more manual calculation
- Contract billing handled automatically — no lookup needed for standard jobs
- Track all fault reports from open to resolved

🔧 Field Engineer

- Receive job assignments on phone with full device and hospital details
- Tap NFC on the device to start — timer begins automatically
- Device history, contract status, and spec PDF available immediately after tap
- Ask the device manual a question in plain English — get the relevant section instantly
- Log parts used — billing status is determined automatically
- Log mileage and meals as you go — receipt photo required
- Tap NFC to end the job — GPS confirmed, record submitted
- Report parts shortages without calling the office
- All of this works with weak hospital WiFi — critical actions queue and sync

🏥 Hospital Staff

- Open the Utiliti link on any phone — no app download needed
- Tap the NFC sticker on the faulty device (Android) or search by device name
- Device details pre-filled — just describe the fault and add photos
- Submit in under 30 seconds
- Check fault report status any time — no more calling Arjo to ask for an update
- See all their hospital's equipment and last service dates

5. Build Plan — 2 Developers, 2 Months

Parallel Streams

With 2 developers the most efficient split is one developer per stream, running concurrently from week 1:

Week	Developer 1 — Back Office	Developer 2 — Mobile & AI
1–2	Database migrations + all Eloquent models + API scaffolding + Auth (Sanctum)	React Native Expo project setup + Auth + API client + NFC read flow (basic)
3–4	Job API (state machine, NFC validation, cost calculation) + Asset API (NFC commissioning) + Contract API + Cost API	Engineer App: Job list, job detail, NFC start/end, parts logging, expense logging
5	Parts + Inventory API + Fault report API + WebSocket events (Laravel Echo)	Engineer App: GPS clock-in/out + geofence logic + push notifications + PDF viewer
6	Google Maps integration (Back Office map, engineer GPS dots, hospital pins, geofence) + Opportunistic PM alert logic	PDF NLP pipeline: FastAPI microservice + vector embeddings + semantic search API + App search UI
7	Back Office Web UI: Dashboard + Job management + Asset management + Map view + Contract management + Fault queue	Hospital mobile web form (fault report + status tracking) + end-to-end integration testing
8	Back Office Web UI: Parts, Reports, System Config, User Management + Performance testing + Bug fixes	NFC hardware testing on real tags + Deployment + SSL + Final QA

Function Count by Stream

Stream	Functions	Count
Back Office API	All business logic, state machine, cost calculation, location engine, WebSocket events	Covers all domains
Back Office Web UI	Dashboard, jobs, assets, contracts, map, parts, faults, reports, system config	All back office pages
Engineer App	Job flow, NFC read + write, expense logging, GPS, PDF viewer, NLP search, push notifications	A-01 to A-05, B-02–B-05, D-01, D-03, E-02, E-03, F-02, H-02
Hospital Web Form	Fault report, device lookup, status tracking	G-01, G-03
AI / Location	Google Maps integration, geofence logic, PM alert, PDF NLP pipeline	F-03, F-04, A-07, D-03

Total Scope Summary

Metric	Count
Total functions in scope	30
Location features (core)	4 (F-03, F-04, A-07, F-02)
Smart beta feature	1 (D-03 — Natural Language Manual Q&A)
Deliberate cuts vs full MVP	4 (hospital native app, NL contract query, contract PDF parsing, cost report export)
User groups served	3 (Back Office, Engineer, Hospital Staff)
Mobile apps	Engineer App (native iOS + Android) + Hospital Web Form
Map provider	Google Maps
Estimated build time	8 weeks / 2 developers (~16 developer-weeks)

6. Feedback Collection Plan

The purpose of this early release is to get Arjo HK using the system and generating structured feedback that drives the G2 priority order. The following questions should be answered by the end of the first 4–6 weeks of live use:

Area	Feedback Questions to Collect	Who to Ask
NFC workflow	Is tapping to start and end jobs reliable on the devices Arjo uses? Are there any hospital environments where NFC read is inconsistent?	Engineers
Job dispatch	Has the time to assign a job decreased? Is the manual assignment process fast enough or is the location recommendation feature needed sooner?	Coordinators
Live map	Is the Google Maps view useful in daily operations? Do the hospital pins and engineer dots provide the visibility coordinators were missing?	Coordinators
Geofence alerts	Are the alerts triggering correctly? Are there false positives? Is the 200m radius appropriate for Arjo HK hospital sites?	Coordinators + Engineers
Cost calculation	Is the NFC-based labour cost accurate? Are expense categories sufficient or are other overhead types needed?	Engineers + Finance
Contract billing	Is the automatic billing flag correct for standard jobs? Are there coverage edge cases the system doesn't handle?	Coordinators + Finance
Manual search Q&A	Are engineers using the natural language search? Which questions come up most often? How accurate are the answers for Arjo's specific devices?	Engineers
Hospital web form	Is the mobile web form adequate or is the lack of a native app causing friction? Is browser NFC scan reliable on Android?	Hospital staff
Parallel running	What is still being done on paper or Excel alongside the system? Why? What would need to change for full replacement?	All users

G2 priority decision: At 4–6 weeks after go-live, a review session with Arjo HK should produce a ranked list of G2 features to build next. The answers to the feedback questions above will determine whether location-aware scheduling recommendation, the native hospital app, or contract PDF auto-parsing is the highest priority addition.

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For questions contact the Utiliti product team. For engineering queries refer to ARCHITECTURE.md.